

Astronomy

ASTRONOMY IS THE SCIENTIFIC STUDY OF STARS AND OTHER OBJECTS IN SPACE.

People have been mapping the stars and tracking the movements of planets for thousands of years.

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The Sun 232–233 ▶

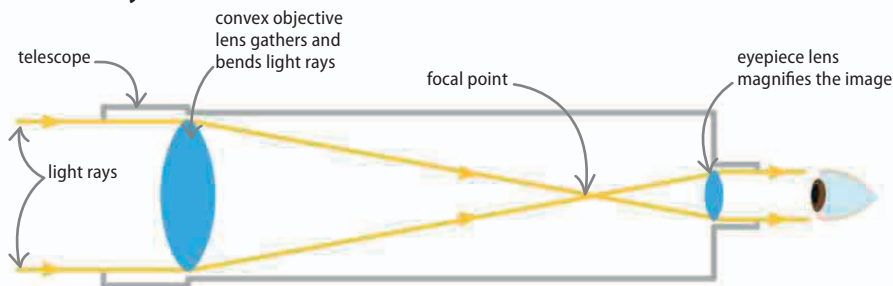
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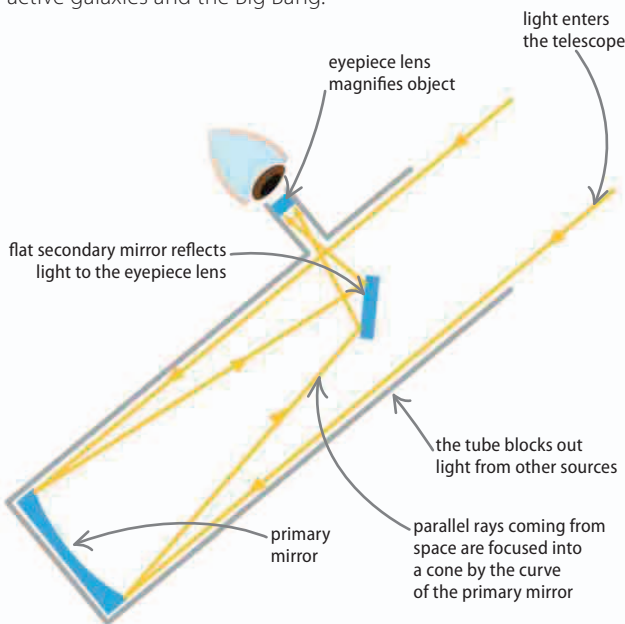
Telescopes

The first telescopes, developed in the early 17th century, gathered light from a distant source and magnified the image using either lenses (in refracting telescopes) or mirrors (in reflecting telescopes). These are called optical telescopes, because they focus light. Today, there also telescopes that reveal other types of radiation invisible to the human eye, such as gamma rays and radio waves. These have led to many important discoveries in astronomy, such as active galaxies and the Big Bang.



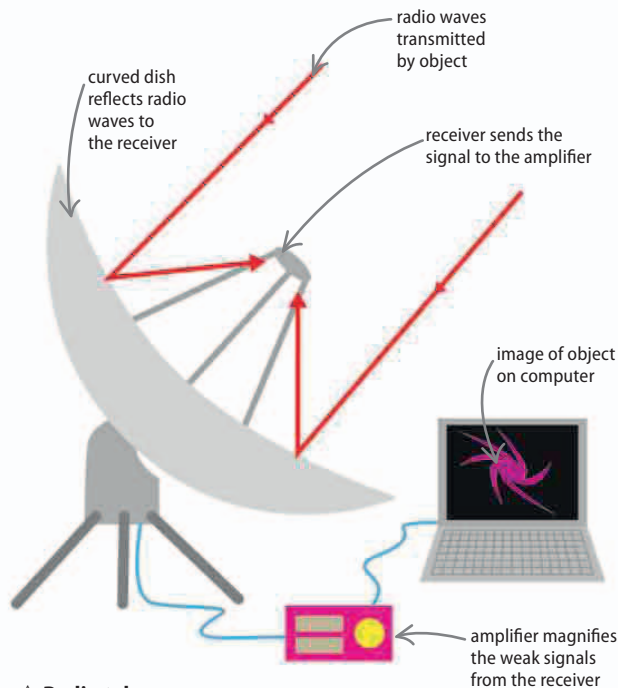
△ Refracting telescope

The large objective lens focuses the rays of light into a small image inside the device. Then an eyepiece lens magnifies the image.



△ Reflecting telescope

This telescope collects light using a curved primary mirror, which reflects and focuses the light onto a flat secondary mirror. This shines the light toward the eyepiece lens, which magnifies the object. The world's most powerful astronomical telescopes are reflecting telescopes, some with mirrors up to 10 m (33 ft) across.



△ Radio telescope

A radio telescope is a huge antenna that picks up the longer wavelengths of radiation coming from space. The radio signals from stars are weak, so a large dish is used to reflect them onto the central receiver. The signals are amplified electronically, and a computer processes them to produce pictures, called radio images.